

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$\mathcal{K}_{\alpha\beta\chi}$	$\mathcal{K}_{\alpha\chi\beta} == \mathcal{K}_{\alpha\beta\chi}$ & $\mathcal{K}_{\beta\alpha\chi} == \mathcal{K}_{\alpha\beta\chi}$	$\begin{aligned} &\mathcal{K}_{3^{-}}^{\#1}{}_{\alpha\beta\chi} + \frac{1}{15} \eta_{\beta\chi} \mathcal{K}_{1^{-}}^{\#2}{}_{\alpha} + \frac{1}{15} \eta_{\alpha\chi} \mathcal{K}_{1^{-}}^{\#2}{}_{\beta} + \frac{1}{15} \eta_{\alpha\beta} \mathcal{K}_{1^{-}}^{\#2}{}_{\chi} + \\ &\frac{1}{9} \eta_{\beta\chi} \mathcal{K}_{0^{+}}^{\#2} n_{\alpha} + \frac{1}{3} \mathcal{K}_{2^{+}}^{\#1}{}_{\beta\chi} n_{\alpha} + \frac{1}{9} \eta_{\alpha\chi} \mathcal{K}_{0^{+}}^{\#2} n_{\beta} + \frac{1}{3} \mathcal{K}_{2^{+}}^{\#1}{}_{\alpha\chi} n_{\beta} - \\ &\frac{1}{15} \mathcal{K}_{1^{-}}^{\#2}{}_{\chi} n_{\alpha} n_{\beta} + \frac{1}{3} \mathcal{K}_{1^{-}}^{\#1}{}_{\chi} n_{\alpha} n_{\beta} + \frac{1}{9} \eta_{\alpha\beta} \mathcal{K}_{0^{+}}^{\#2} n_{\chi} + \frac{1}{3} \mathcal{K}_{2^{+}}^{\#1}{}_{\alpha\beta} n_{\chi} - \\ &\frac{1}{15} \mathcal{K}_{1^{-}}^{\#2}{}_{\beta} n_{\alpha} n_{\chi} + \frac{1}{3} \mathcal{K}_{1^{-}}^{\#1}{}_{\beta} n_{\alpha} n_{\chi} - \frac{1}{15} \mathcal{K}_{1^{-}}^{\#2}{}_{\alpha} n_{\beta} n_{\chi} + \\ &\frac{1}{3} \mathcal{K}_{1^{-}}^{\#1}{}_{\alpha} n_{\beta} n_{\chi} - \frac{1}{3} \mathcal{K}_{0^{+}}^{\#2} n_{\alpha} n_{\beta} n_{\chi} + \mathcal{K}_{0^{+}}^{\#1} n_{\alpha} n_{\beta} n_{\chi} \end{aligned}$	$\mathcal{T}_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$\mathcal{K}_{0^{+}}^{\#1}$	(None)	$\mathcal{K}_{\alpha\beta\chi} n^{\alpha} n^{\beta} n^{\chi}$	$\mathcal{T}_{0^{+}}^{\#1}$
$\mathcal{K}_{0^{+}}^{\#2}$	(None)	$3 \mathcal{K}_{\alpha\beta}^{\beta} n^{\alpha} - 3 \mathcal{K}_{\alpha\beta\chi} n^{\alpha} n^{\beta} n^{\chi}$	$\mathcal{T}_{0^{+}}^{\#2}$
$\mathcal{K}_{1^{-}}^{\#1}{}_{\alpha}$	(None)	$3 \mathcal{K}_{\alpha\beta\chi} n^{\beta} n^{\chi} - 3 \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\beta} n^{\chi} n^{\delta}$	$\mathcal{T}_{1^{-}}^{\#1}{}_{\alpha}$
$\mathcal{K}_{1^{-}}^{\#2}{}_{\alpha}$	(None)	$3 \mathcal{K}_{\alpha\beta}^{\beta} - 3 \mathcal{K}_{\beta\chi}^{\chi} n_{\alpha} n^{\beta} - 3 \mathcal{K}_{\alpha\beta\chi} n^{\beta} n^{\chi} + 3 \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\beta} n^{\chi} n^{\delta}$	$\mathcal{T}_{1^{-}}^{\#2}{}_{\alpha}$
$\mathcal{K}_{2^{+}}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^{+}}^{\#1}{}_{\beta\alpha} == \mathcal{K}_{2^{+}}^{\#1}{}_{\alpha\beta}$	$\begin{aligned} &3 \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \eta_{\alpha\beta} \mathcal{K}_{\chi}^{\delta}{}_{\delta} n^{\chi} + \mathcal{K}_{\chi}^{\delta}{}_{\delta} n_{\alpha} n_{\beta} n^{\chi} - 3 \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} - \\ &3 \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta} + \eta_{\alpha\beta} \mathcal{K}_{\chi\delta\epsilon} n^{\chi} n^{\delta} n^{\epsilon} + 2 \mathcal{K}_{\chi\delta\epsilon} n_{\alpha} n_{\beta} n^{\chi} n^{\delta} n^{\epsilon} \end{aligned}$	$\mathcal{T}_{2^{+}}^{\#1}{}_{\alpha\beta}$
$\mathcal{K}_{3^{-}}^{\#1}{}_{\alpha\beta\chi}$	$\mathcal{K}_{3^{-}}^{\#1}{}_{\alpha\chi\beta} == \mathcal{K}_{3^{-}}^{\#1}{}_{\alpha\beta\chi}$ & $\mathcal{K}_{3^{-}}^{\#1}{}_{\beta\alpha\chi} == \mathcal{K}_{3^{-}}^{\#1}{}_{\alpha\beta\chi}$	$\begin{aligned} &\mathcal{K}_{\alpha\beta\chi} - \frac{1}{5} \eta_{\beta\chi} \mathcal{K}_{\alpha}^{\delta}{}_{\delta} - \frac{1}{5} \eta_{\alpha\chi} \mathcal{K}_{\beta}^{\delta}{}_{\delta} - \frac{1}{5} \eta_{\alpha\beta} \mathcal{K}_{\chi}^{\delta}{}_{\delta} + \frac{1}{5} \mathcal{K}_{\chi}^{\delta}{}_{\delta} n_{\alpha} n_{\beta} + \\ &\frac{1}{5} \mathcal{K}_{\beta}^{\delta}{}_{\delta} n_{\alpha} n_{\chi} + \frac{1}{5} \mathcal{K}_{\alpha}^{\delta}{}_{\delta} n_{\beta} n_{\chi} - \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{5} \eta_{\beta\chi} \mathcal{K}_{\delta}^{\epsilon}{}_{\epsilon} n_{\alpha} n^{\delta} - \\ &\mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} + \frac{1}{5} \eta_{\alpha\chi} \mathcal{K}_{\delta}^{\epsilon}{}_{\epsilon} n_{\beta} n^{\delta} - \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{5} \eta_{\alpha\beta} \mathcal{K}_{\delta}^{\epsilon}{}_{\epsilon} n_{\chi} n^{\delta} - \\ &\frac{3}{5} \mathcal{K}_{\delta}^{\epsilon}{}_{\epsilon} n_{\alpha} n_{\beta} n_{\chi} n^{\delta} + \frac{1}{5} \eta_{\beta\chi} \mathcal{K}_{\alpha\delta\epsilon} n^{\delta} n^{\epsilon} + \frac{1}{5} \eta_{\alpha\chi} \mathcal{K}_{\beta\delta\epsilon} n^{\delta} n^{\epsilon} + \\ &\frac{1}{5} \eta_{\alpha\beta} \mathcal{K}_{\chi\delta\epsilon} n^{\delta} n^{\epsilon} + \frac{4}{5} \mathcal{K}_{\chi\delta\epsilon} n_{\alpha} n_{\beta} n^{\delta} n^{\epsilon} + \frac{4}{5} \mathcal{K}_{\beta\delta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \\ &\frac{4}{5} \mathcal{K}_{\alpha\delta\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon} - \frac{1}{5} \eta_{\beta\chi} \mathcal{K}_{\delta\epsilon\phi} n_{\alpha} n^{\delta} n^{\epsilon} n^{\phi} - \frac{1}{5} \eta_{\alpha\chi} \mathcal{K}_{\delta\epsilon\phi} n_{\beta} n^{\delta} n^{\epsilon} n^{\phi} - \\ &\frac{1}{5} \eta_{\alpha\beta} \mathcal{K}_{\delta\epsilon\phi} n_{\chi} n^{\delta} n^{\epsilon} n^{\phi} - \frac{2}{5} \mathcal{K}_{\delta\epsilon\phi} n_{\alpha} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon} n^{\phi} \end{aligned}$	$\mathcal{T}_{3^{-}}^{\#1}{}_{\alpha\beta\chi}$